$\Rightarrow \alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n - \alpha_{n+1} u_{n+1} - \dots - \alpha_p u_p = 0$

$\Rightarrow \alpha_i = 0 \quad , i = 1, \dots, p \quad [\because \{u_1, u_2, \dots, u_p\}$ is the basis of $U]$

$\Rightarrow A$ is LI.

Now, it remains to show that $R(T) = [A]$.

$$R(T) = [\{T(u_1), T(u_2), \dots, T(u_n), \dots, T(u_p)\}]$$

$$T(u_i) = 0 \quad , i = 1, \dots, n$$

$$R(T) = [\{T(u_{n+1}), \dots, T(u_p)\}]$$

$$= [A].$$

*Q:- $T : \mathbb{R}^4 \longrightarrow \mathbb{R}^3$

$T(e_1) = (1,1,1)$

$T(e_2) = (1,-1,1)$

$T(e_3) = (1,0,0)$

$T(e_4) = (1,0,1)$

HW Find $T(x,y,z,t) = ?$

$$R(T) = [\{(1,1,1), (1,-1,1), (1,0,0), (1,0,1)\}]$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1 \atop R_4 \to R_4 - R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 0 \end{pmatrix} \xrightarrow{R_4 \to 2R_4 - R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Pkalika Maths